My own test rejected my best method.

It couldn’t see a win that size.

I built an open-source benchmark: seven ways to build a portfolio, 18 US ETFs, 168 months, after trading costs, one rule fixed before the run. Then I re-read the result with the test the design was built for, and ran it on a market I had never seen.



The test’s 80% detection threshold, in net excess Sharpe: return above cash per unit of risk, after trading costs. An improvement at the bar would usually be missed.

What this is

A benchmark for portfolio methods: seven constructions, one decision rule, and a test suite that checks the evaluation itself. Not a trading strategy, and not investment advice.

Why it matters

A model can be built correctly and still be judged by a test that cannot see its effect. That is the failure this project exists to catch — and the first one it caught was mine.

The edge didn’t survive a new market. The lesson did.

THE PROBLEM

Portfolio construction is an estimation problem.

The standard method estimates average returns and correlations from history, then picks the portfolio with the best trade-off between return and risk. The estimates are noisy. The optimizer can load on assets that looked good by chance. With 18 assets and 60 months of history, there are more parameters than independent observations; regularization — shrinking noisy estimates — is the standard response.

The project asks a narrow question: do "robust" methods — a shrinkage covariance (which pulls noisy correlations toward a simpler estimate), a Black–Litterman mean (which blends the historical average with a prior) — beat a plain baseline out of sample, after costs, once you account for having tried more than one method?

Why it matters

A better estimator is only useful if it survives the noise, the costs, and the decision. That is what the benchmark tests.

This is a question about evidence, not a pitch for a strategy.

A benchmark, not a backtest.

A backtest is a model of a market. A benchmark tests whether the model's edge survives data it has not seen — and whether the test could see the edge at all.

The engine	A walk-forward test: re-estimate each month on the previous 60 months, then test the next month. 168 out-of-sample months in 14 non-overlapping folds.
The universe	18 US ETFs across equity, bonds, commodities and real estate; eligibility rules applied at each date.
The methods	Five standard baselines plus two variants that change one assumption in the standard mean-variance model.
The apparatus	A pre-registered decision rule, a leakage test suite, a numerical check that the optimizer solved the problem, and a reproduction check.

1176 optimizer solves; every one checked numerically. The code and the frozen data snapshot are public.

What it is not: a live system, a forecast, or a claim of alpha. One asset family, one period, flat costs.

The benchmark is the deliverable, not just the headline number.

One metric, one rule, fixed before the run.

Net excess Sharpe: return above cash, after 10 bps (0.10%) one-way trading costs, divided by volatility. A summary of the path, not a forecast.

The rule: a method had to beat the best baseline by at least 0.20 net Sharpe, clear the significance tests (the edge is unlikely to be luck), and pass five more checks on turnover, drawdown, cost levels, folds and concentration.

Why both tests

Significance alone admits tiny effects. Size alone admits noise. Requiring both, plus robustness, is what makes a null informative.

The rule was written before the run. The run is exploratory because the planned blind test was compromised.

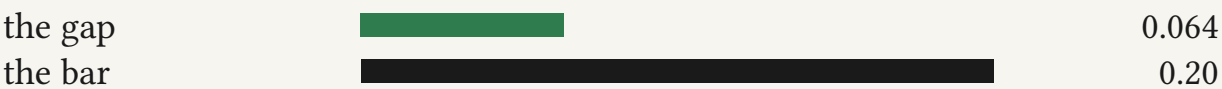
THE RESULT

First on the metric. Failed on the decision.

0.8538

net excess Sharpe — the highest of the seven

Gap over the best baseline	+0.064
Materiality bar	0.20
Original test (probability edge is real)	0.587 of 0.95
Checks failed	3 of 8



A ranking and a verdict are different objects.

The number is a summary of one path, not a return you could spend.

The bar was 0.20. The test needed 0.72.

A test has a detection threshold: the smallest true effect it can reliably notice. This one needed about 0.72 to have an 80% chance of noticing it. The bar was 0.20. A real 0.20 improvement would have been missed most of the time.



What went wrong

The original test compared the candidate's Sharpe with the baseline's observed Sharpe. The design was built for a paired comparison. The test was measuring a different question.

A test can be rigorous and still answer the wrong question.

THE CORRECTION

The same data, the right test.

0.917

probability M2 beats B3 on a paired resampling test

Sharpe gap	+0.064
90% range	−0.011 to 0.147
p-value	0.083
Original test	0.587

The two strategies hold almost the same portfolio (ρ = 0.986), so comparing them month by month is far more sensitive than comparing each to zero.

Pairing is what turns a noisy comparison into a usable one.

THE READING

The data exclude a 0.20 advantage.

0.147 < 0.20

the 90% upper bound is below the materiality bar

NOT MATERIAL

M2 is plausibly a little better. It is not materially better — not enough to justify the extra complexity. The data exclude the 0.20 improvement the study was designed to detect.

This is a bounded result: material, not material, or undecided. Here it is not material.

The result is a bound, not a shrug.

WHY IT FAILED

The cap did not rescue shrinkage.

I thought the 30% position cap was doing the job of keeping the portfolio simple, which would explain why the shrinkage method added nothing. I tested it.

Position limit	Shrinkage	Black–Litterman
0.20	−0.015 (−0.107, 0.073)	+0.049 (−0.031, 0.140)
0.30	−0.004 (−0.096, 0.080)	+0.064 (−0.011, 0.147)
0.40	+0.010 (−0.084, 0.097)	+0.119 (0.037, 0.206)
1.00	−0.003 (−0.095, 0.083)	+0.138 (0.044, 0.237)

Shrinkage: not supported. The gap stays at or below zero; every interval includes zero.

Black–Litterman: depends on the cap. The gap grows as the cap loosens, but remains below 0.20. The looser cap also means larger positions: the largest holding rises from 0.30 to 0.80.

Post-hoc, and reported as post-hoc: this tests an explanation for the null, not a new hypothesis.

The mechanism test did not support the shrinkage explanation. It did show the cap binding the mean treatment.

THE NEW SAMPLE

A new market. A different winner.

18 country ETFs · 2005–2026 · 252 months · 21 folds. The best baseline was equal weight (B1), not sample MVO (B3).

Method	Net Sharpe	Turnover
B1	0.3527	0.83
B2	0.3423	0.86
M2	0.3193	2.48
B5	0.2919	1.40
B3	0.2797	2.54
B4	0.2715	1.76
M1	0.2710	2.43

B1 equal weight · B2 risk parity · B3 sample MVO · B4 minimum variance · B5 turnover-penalised · M1 shrinkage · M2 Black–Litterman.

M2 vs B1: −0.033 [−0.125, 0.075]. **M1 vs B1:** −0.082 [−0.175, 0.029]. Both upper bounds are below 0.20 both won 0 of 5 cost levels; both traded about 3.0× the baseline.

What replicated — and what did not

The benchmark lesson replicated: the best baseline changed from B3 to B1. The treatment advantage did not.

The edge depended on the market. The lesson did not.

The benchmark is the binding constraint.

What you compare against decides the verdict. The metric is a summary; the benchmark is the decision.

First sample: the top number failed the rule — +0.064 against 0.20.

Second sample: equal weight beat both treatments.

The correction: the same data, read with the test the design was built for.

The limit: every number here belongs to its sample, not to markets in general.

What I would carry into the next study

— Fix the decision rule before the data.

— Audit the test's power, not only the strategy's return.

— Replicate on a sample that did not inform the design.

Code, paper and the reproduction check: github.com/kira-ml/robustfolio